

Characteristics of Logic Gates [Logic gates are basic building block of any digital system]

1. A logic gate can have multiple input but will only have one output.
2. The relationship between input and output is based on certain logic.
3. The gates are named based on the logic.

Basic Gates

- ① Not gate
- ② AND gate
- ③ OR gate

Universal Gates

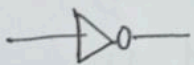
- ① NAND Gate
- ② NOR Gate

Exclusive Logic Gates

- ① XOR gate
- ② XNOR gate

NOT Gate

Symbol

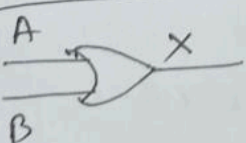


Truth Table

Input	Output
0	1
1	0

Output is opposite of the input

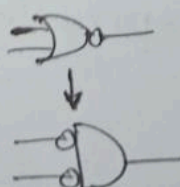
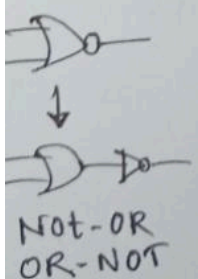
OR Gate



A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

O/p high if one or all i/p high

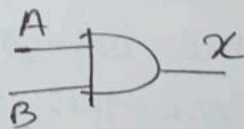
NOR Gate



Negative AND

O/P of NOR is low one or more or all i/p are high

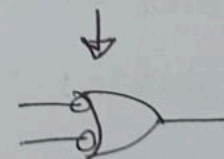
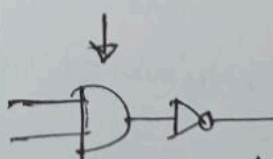
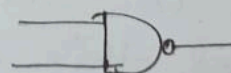
AND Gate



A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

O/P high when all inputs are high

NAND Gate



Not-AND/
AND-Not

Negative OR

Invert After
AND Operation

Invert individual
i/p Before OR
Operation

A	B	AND	Not
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

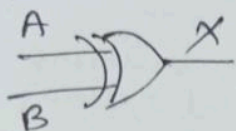
Both Gives Same

A	B	Invert	OR
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

O/p high one or all i/p are low

Bubble
shifting
if bubble
cross the
Gate
AND → OR
OR → AND

XOR Gate

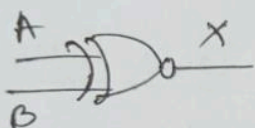


Output of XOR gate is high whenever the two inputs are different. If same the output is low.

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

$X = AB' + A'B$ ← we will learn to derive the expression in next classes.

XNOR Gate



The output of XNOR Gate is high whenever the two inputs are same

A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

$$X = AB + A'B'$$

Boolean Algebra

Boolean algebra is the category of ~~all~~ Algebra in which the variable values are the truth values (true/false) ordinarily denoted by 1 for true, 0 for false. It is the mathematics of digital logic. Boolean algebra is used to simplify the boolean expressions or Boolean function. Boolean Algebra is also used in Set theory and Statistics. George Boole established this algebra of logic.

Boolean Variable: The variables can either take the value 1 or 0.

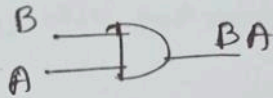
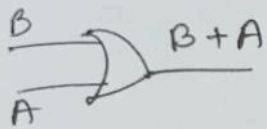
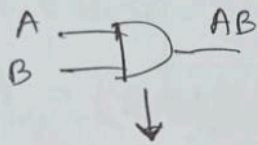
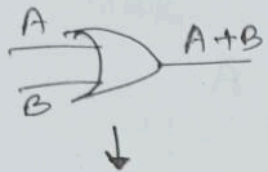
Boolean Logic Expression: The expression is comprised of only Boolean variables. The output of expression is a Boolean value, either zero or one.

Example: $AB + A(B+C) + B(B+C)$

15 of Boolean Algebra

* The commutative law [अविवर्तनधर्म] ज्ञान अविवर्तन कणी

Addition: $A+B = B+A$ Multiplication: $AB = BA$



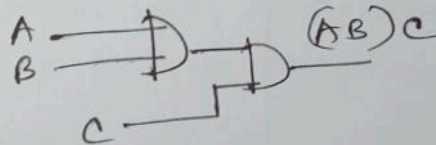
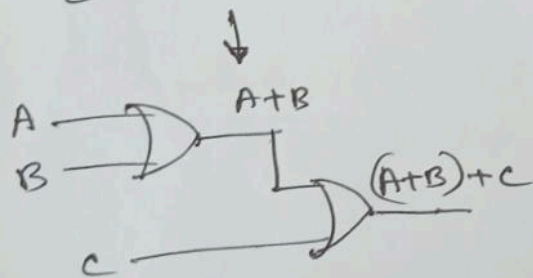
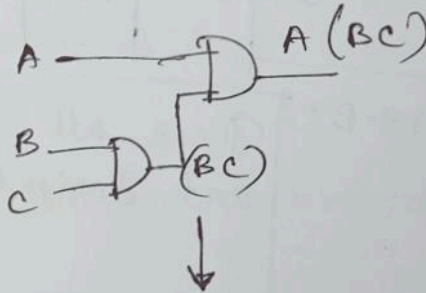
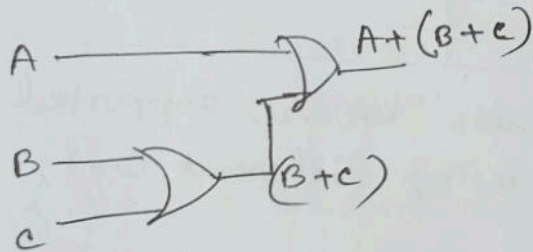
* Associative Law [अश्यामी] अश्यामी कणन

Addition

$$A+(B+C) = (A+B)+C$$

Multiplication

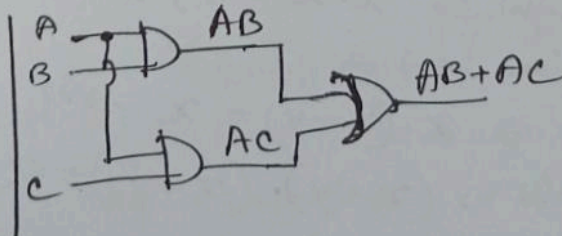
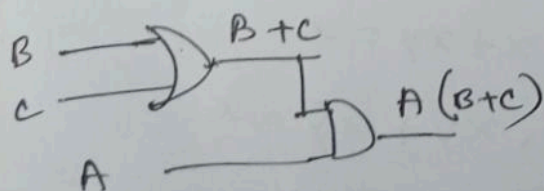
$$A(BC) = (AB)C$$



* Distributive Law [विभक्तन कणी]

The distributive law for three variable is written as

$$A(B+C) = AB+AC$$



Rules of Boolean Algebra

Basic Rules of Boolean Algebra

$$1. A + 0 = A$$

$$2. A + 1 = 1$$

$$3. A \cdot 0 = 0$$

$$4. A \cdot 1 = A$$

$$5. A + A = A$$

$$6. A + \bar{A} = 1$$

$$7. A \cdot A = A$$

$$8. A \cdot \bar{A} = 0$$

$$9. \bar{\bar{A}} = A$$

$$10. A + AB = A \rightarrow A(1+B) = A \cdot 1 = A$$

Rule 2
↓
Rule 4

$$11. A + \bar{A}B = A + B$$

How??

A	B	$\bar{A}$	$\bar{A}B$	$A + \bar{A}B$	$A + B$
0	0	1	0	0	0
0	1	1	1	1	1
1	0	0	0	1	1
1	1	0	0	1	1

sounds good??

12.

$$(A+B)(A+C) = A + BC$$

These all Rules are supported and derived using different law.

$$\Rightarrow A \cdot A + A \cdot C + A \cdot B + B \cdot C$$

$$\Rightarrow A(1+C+B) + BC$$

$$\Rightarrow A \cdot 1 + BC$$

↑ 2nd Rule

$$\Rightarrow A + BC$$

$$1. \text{Annulment Law} \Rightarrow x \cdot 0 = 0, x + 1 = 1$$

$$2. \text{Identity Law} \Rightarrow x \cdot 1 = x, x + 0 = x$$

$$3. \text{Idempotent Law} \Rightarrow x \cdot x = x, x + x = x$$

$$4. \text{Complement Law} \Rightarrow x \cdot \bar{x} = 0, x + \bar{x} = 1$$

$$5. \text{Double Negation Law} \Rightarrow \bar{\bar{x}} = x$$

$$6. \text{Absorption Law} \Rightarrow x \cdot (x + y) = x \rightarrow ?? \quad x + xy = x$$

$$7. \text{Redundancy Law} \Rightarrow (x + y) \cdot (x + \bar{y}) = x, xy + x\bar{y} = x$$

$$(x + \bar{y}) \cdot y = xy, x\bar{y} + y = x + y$$

$$8. \text{Consensus Law} \Rightarrow (x + y) \cdot (\bar{x} + z) \cdot (y + z) = (x + y) \cdot (\bar{x} + z)$$

$$xy + \bar{x}z + yz = xy + \bar{x}z$$

$$9. \text{XOR} \Rightarrow x \oplus y \Rightarrow (x + \bar{y}) \cdot (\bar{x} + y) \Rightarrow \bar{x}y + x\bar{y}$$

$$10. \text{XNOR} \Rightarrow x \odot y \Rightarrow (x + y) \cdot (\bar{x} \cdot \bar{y}) \Rightarrow \bar{x}\bar{y} + xy \Rightarrow (x + y) \cdot (\bar{x} + \bar{y})$$

Theorem: A theorem is a mathematical statement that has been proven on the basis of previously established statements, often using other theorems. Famous Pythagorean theorem, which has nearly 400 proofs.

Theory: Theories explain natural phenomena using scientific method, testing, confirmation, observation and experimentation. Theory of Relativity.

Fact: A true statement. Sun Rises in the East

Law: A statement based on repeated observations. It express relationships between entities under certain conditions, and often expressed mathematically.

Proof: A successful demonstration of a hypothesis under study using evidence and analysis.

Rule: A statement that express a philosophical point.

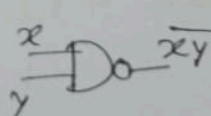
hypothesis: A proposed explanation made on the basis of limited evidence as a starting point of further investigation.

De Morgans Theorem

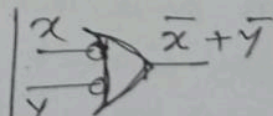
Augustus De Morgan stated ~~that~~ two theorems

Theorem 1 ① The complement of a product of variables is equal to the sum of the ^{individual} complements of variables

$$\overline{xy} = \bar{x} + \bar{y}$$



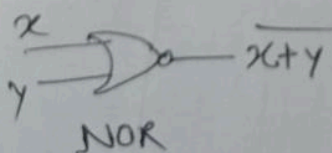
NAND



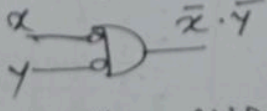
Negative-OR

Theorem 2 ② The complement of a sum of variables is equal to the product of the individual complement of the variables

$$\overline{x+y} = \bar{x} \cdot \bar{y}$$



NOR



Negative-AND

$$\cdot \& - \Rightarrow +$$

$$+ \& - \Rightarrow \cdot$$

Application of All these Laws, Rules and Theorems

$$\overline{xyz} = \bar{x} + \bar{y} + \bar{z} ; \overline{x+y+z} = \bar{x} \cdot \bar{y} \cdot \bar{z} = \overline{xyz}$$

$$(A) \overline{(A+B+C)D} = (\bar{A} \cdot \bar{B} \cdot \bar{C}) + \bar{D}$$

$$\Rightarrow \overline{(AD+BD+CD)}$$

$$\Rightarrow (\bar{A} + \bar{D})(\bar{B} + \bar{D})(\bar{C} + \bar{D}) \rightarrow \text{you can take it further}$$

$$(b) \overline{ABC + DEF} = (\bar{A} + \bar{B} + \bar{C})(\bar{D} + \bar{E} + \bar{F})$$

$$(c) \overline{AB + CD + EF} = (\bar{A} + \bar{B})(\bar{C} + \bar{D})(\bar{E} + \bar{F})$$

Take further

* Apply Boolean ^{Theorem} Expressions in Following Expressions

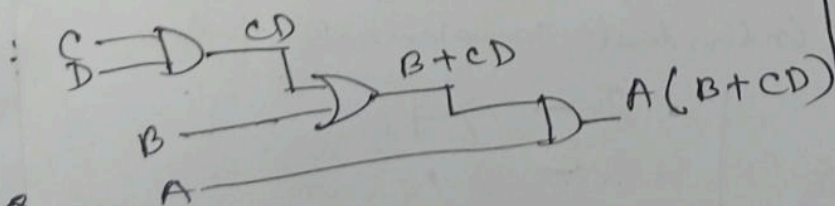
$$(a) \overline{ABC + (\bar{D} + E)} \quad (b) \overline{(A+B)C} \quad (c) \overline{A+B+C + \bar{D}E}$$

Boolean Application in Logic Circuits

Boolean algebra provides a concise way to express the operation of a logic circuit formed by a combination of Logic gates so that the output can be determined for various combinations of input values

→ A logic circuit can be expressed by Boolean expression and Boolean expression can be implemented by a logic circuit.

$$A(B+CD)$$



A	B	C	D	CD	B+CD	A(B+CD)
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	1	1	0
0	1	0	0	0	1	0
0	1	0	1	0	1	0
0	1	1	0	0	1	0
0	1	1	1	1	1	0
1	0	0	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	0	0	0
1	0	1	1	1	1	1
1	1	0	0	0	1	1
1	1	0	1	0	1	1
1	1	1	0	0	1	1
1	1	1	1	1	1	1

8

9

10

Logic Simplification and its Importance

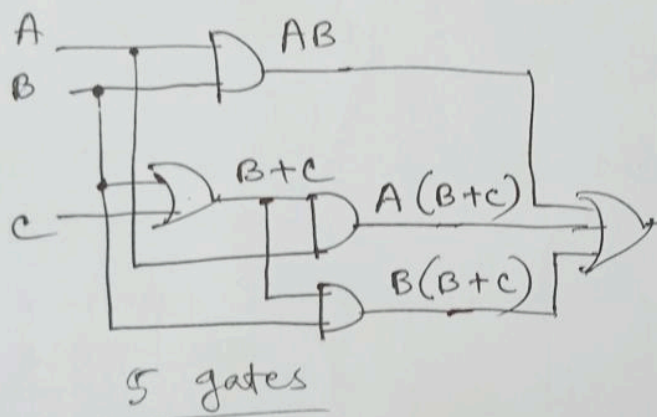
* A large Boolean expression can often be simplified to a simpler and shorter expression.

* This is done by applying Laws and Rules of Boolean Algebra.

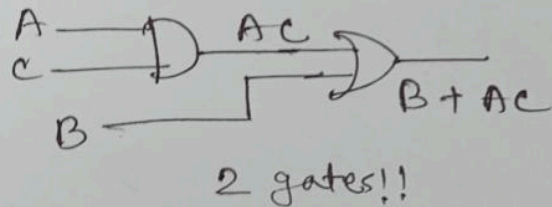
* Simplifying makes implementation ~~simpler~~ easier with lesser number of gates

$$AB + A(B+C) + B(B+C) \Rightarrow \underline{AB} + \underline{AB} + AC + B \cdot B + BC$$

without simplification



$$\begin{aligned} &= \underline{AB} + AC + \underline{B} + BC \\ &= B(\underline{A+1}) + AC + BC \\ &= \underline{B} + AC + \underline{BC} \\ &= B + BC + AC \\ &= B(\underline{1+C}) + AC \\ &= B + AC \end{aligned}$$



Example

$$\underline{\bar{A}BC} + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + \underline{ABC}$$

$$= BC(\bar{A}+A) + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C$$

$$= BC \cdot 1 + \bar{B}\bar{C}(\underline{A+\bar{A}}) + A\bar{B}C$$

$$= \underline{BC} + \bar{B}\bar{C} + \underline{A\bar{B}C}$$

$$= \underline{BC} + \bar{B}(\bar{C} + AC) \quad \text{if you go there there will be no law to Apply}$$

$$= C(B + A\bar{B}) + \bar{B}\bar{C}$$

$$= C(\underline{A\bar{B} + B}) + \bar{B}\bar{C}$$

$x\bar{y} + y$ — Redundancy law

$$= C(A+B) + \bar{B}\bar{C} \Rightarrow AC + BC + \bar{B}\bar{C}$$

Standard Forms of Boolean Expressions

All Boolean expressions can be converted into two standard form. The Sum of Product (SOP) form & Product of Sum (POS) form.

SOP = min-term

POS = Max-term

- Each variable in a Boolean expression is a literal.
- Boolean variables can appear in normal (x) or complement form (x')
- Each AND combination of terms is a minterm
- Each OR combination of terms is a maxterm.
- The resultant of minterm must be 1
- The resultant of maxterm must be 0

SOP

When two or more product terms are summed by Boolean addition the resulting expression is a Sum-of-Products (SOP).

$$AB + AB\bar{C}, \quad ABC + CDE + \bar{B}\bar{C}\bar{D}, \quad \bar{A}B + \bar{A}\bar{B}\bar{C} + AC$$

POS

When two or more sum terms are multiplied, the resulting expression is a product-of-sum (POS).

$$(\bar{A} + B)(A + \bar{B} + C), \quad (\bar{A} + \bar{B} + \bar{C})(C + \bar{D} + E)(\bar{B} + C + D),$$

The standard SOP Form

A standard expression is one in which all the variables in the domain appear in each term in the expression.

$\bar{A}\bar{B}CD + \bar{A}\bar{B}\bar{C}\bar{D} + AB\bar{C}\bar{D}$ is a standard SOP expression.

~~Standard~~ Standard SOP expressions are important to construct truth table. These are also called canonical form

We can convert any non standard SOP expression into standard SOP expression using Boolean algebra ($A + \bar{A} = 1$)

$$\bar{A}\bar{B}C + \bar{A}\bar{B} + AB\bar{C}D$$

The Karnaugh Map

A Karnaugh map provides a systematic method for simplifying Boolean expressions and if properly used, it will produce the simplest SOP or POS expression possible, known as the minimum expression. Karnaugh map provides a cookbook method for simplification. There may be 2, 3, 4, 5 variable Karnaugh. However, we will find 3 and 4 variable Karnaugh Maps as very common.

3 variable K-Map

It is an array of 8 cells.

AB \ C	0	1
00		
01		
11		
10		

4 Variable K-Map

It is an array of 16 cells

AB \ CD	00	01	11	10
00				
01				
11				
10				

Why 11 is placed before 10? It is due to cell adjacency.

Cell Adjacency: The cells in a Karnaugh map are arranged in such a way so that there is only a single-variable change between adjacent cells.

- * Adjacency is defined by a single-variable change.
- * Physically each cell is adjacent to the cells that are immediately next to it on any of the four sides.
- * A cell is not adjacent to the cells that diagonally touch any of its corners.
- * The cells in the top row are adjacent to the cells in the bottom row.
- * The cells in the left column is adjacent to the cells in the right column.
- * This is called "wrap-around" adjacency.

Truth Table to K-Map

* From the following Truth Table form a K-Map

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

A \ B		C	
		0	1
0	0	0	0
0	1	1	1
1	1	0	0
1	0	1	1

Forming the Groups.

To find the minimal SOP/simplified SOP expression we must first group the 1s that are adjacent.

→ To Form the 1s group we should follow these rules.

- ① A group can contain number of cells that is only a power $2(2^n)$ i.e. 1, 2, 4, 8 and 16.
- ② Each cell in a group must be adjacent to some other cell in the group, but not all cells need to be adjacent.
- ③ We have to maximize the number of adjacent cell while minimizing the number of groups.
- ④ Each 1 in the map must be included in at least one group keeping accordance to the rules above.
- ⑤ The 1s included in a group can be included in another group if the overlapping group includes ungrouped 1s keeping accordance to rules 1 to 4.

Truth Table to K-map Example

(2)

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	0	1	1	1
10	1	1	0	0

let's see an example of Forming Groups

Wrap around

AB \ CD	00	01
00	1	
01		1
11	1	1
10		

AB \ CD	00	01	11	10
00	1	1		
01	1			
11			1	
10	1	1		

Wrap around

AB \ CD	00	01	11	10
00	1			1
01	1	1		1
11	1	1		1
10	1		1	1

AB \ CD	00	01	11	10
00	1	1		
01	1	1	1	1
11				
10		1	1	

Finding the minimal SOP form

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	1	0	1
11	1	1	0	1
10	1	1	1	1

Similarly vertically for this group.

unchanged
01 → 11

changed so we will discard this variable. we will take B from here. So, the expression will be $B\bar{C}$.

$\bar{D}$

For CD, everything gets changed, so, we will not take anything.

the expression will be $A\bar{B}$

from the last group we will take $\bar{D}$. Because the D variable remains unchanged in 0 form, vertically AB variables gets changed while moving to downward. So, we do not take anything from AB.

So, the minimal expression (SOP) =

$$F = A\bar{B} + B\bar{C} + \bar{D}$$

Move along a group row wise and column wise step by step.

→ $B\bar{C}$ ⇒ The group is two cell wide two cell long. Move horizontally. You can see the variable D get's changed from 0 → 1. So we discard it. Which is unchanged? C, in which form? $\bar{C}$ → means 0.

Mapping a Standard SOP Expression.

(3)

$$\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C$$

0 0 0 0 0 1 1 1 0 1 0 0

$$\begin{aligned} &0011 \quad 0100 \quad 1101 \quad 1111 \quad 1100 \\ &\bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}CD + A\bar{B}\bar{C}\bar{D} \\ &\quad + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} \\ &\quad 0001 \quad 1010 \end{aligned}$$

AB \ C	0	1
00	1	1
01		
11	1	
10	1	

AB \ CD	00	01	11	10
00		1	1	
01	1			
11	1	1	1	
10				1

Mapping a Non standard SOP Expression

$$\bar{A} + A\bar{B} + A\bar{B}\bar{C} \quad \left[\text{This is a 3 variable Expression} \right]$$

only $A\bar{B}\bar{C}$ is completed form.

A	B	C	
0	0	0	$\bar{A}\bar{B}\bar{C}$
0	0	1	$\bar{A}\bar{B}C$
0	1	0	$\bar{A}B\bar{C}$
0	1	1	$\bar{A}BC$
1	0	0	$A\bar{B}\bar{C}$
1	0	1	$A\bar{B}C$
1	1	0	$AB\bar{C}$
1	1	1	ABC

Now see how many combination we can get that are having $\bar{A}$

$$\bar{A}\bar{B}\bar{C}, \bar{A}\bar{B}C, \bar{A}B\bar{C}, \bar{A}BC$$

Similarly look for the combinations that contains $A\bar{B}$, $\rightarrow A\bar{B}\bar{C}, A\bar{B}C$

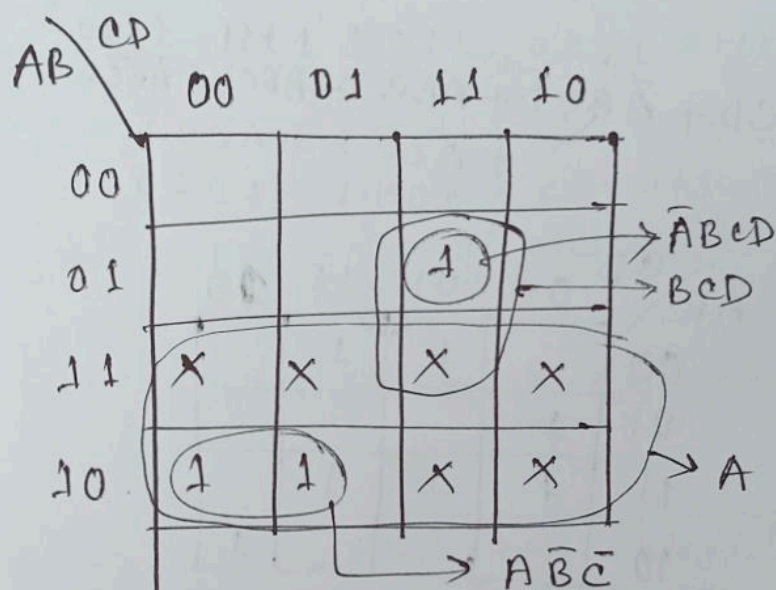
So, the K-map will be

AB \ C	0	1
00	1	1
01	1	1
11	1	
10	1	1

Don't Care Condition

Sometimes a situation arises in which some input variable combinations are not allowed. For example, in BCD code there are six invalid combinations.

	A	B	C	D	Output
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	0
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	0
6	0	1	1	0	0
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	1
A	1	0	1	0	X
B	1	0	1	1	X
C	1	1	0	0	X
D	1	1	0	1	X
E	1	1	1	0	X
F	1	1	1	1	X



without Don't care

$$Y = A\bar{B}\bar{C} + \bar{A}BCD$$

with Don't care

$$Y = A + BCD$$

K-map (POS minimization)

$$(A + B + C)(A + \bar{B} + C)(\bar{A} + \bar{B} + C)(\bar{A} + B + \bar{C})$$

$$(0 \ 0 \ 0)(0 \ 1 \ 0)(1 \ 1 \ 0)(1 \ 0 \ 1)$$

